

“Flower and the Youth”: Virtual Narrative and Creative Methods of Oral Performance in China Intangible Cultural Heritage

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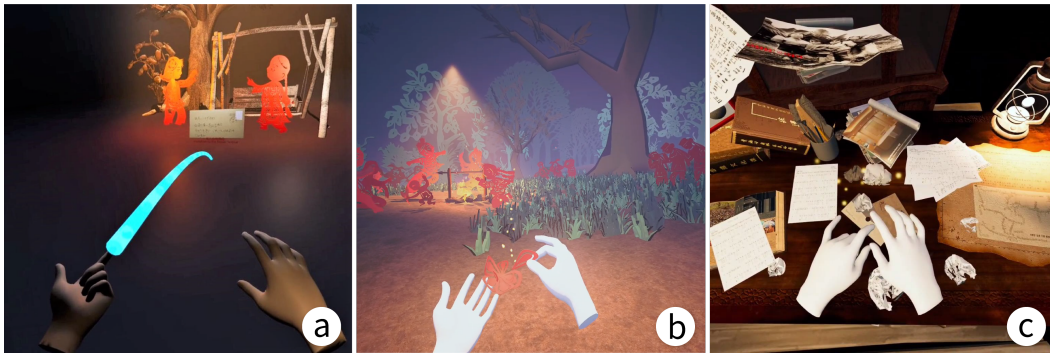


Figure 1: Interactive performance narrative based on gesture recognition. (a) Hand gesture for teleporting. (b) Hand gesture for collecting and grabbing petals. (c) Hand gesture for manipulating objects.

ABSTRACT

The intangible cultural heritage of China contains many different forms of performing arts, of which oral performance is an important branch. “Hua’er” is the most popular folk performance sung in the Hui ethnic area of Ningxia. Based on technology of virtual reality (VR) and gesture recognition, our work proposes three design methods: interactive performance narrative, metaphorical elements and embodied cognition, applied to the VR performance, “Flower and the Youth”. VR can provide audience with a more immersed experience that contributes to the transmission and dissemination of non-heritage performing arts. Our work provides universal design approaches to the creation of content for future intangible cultural heritage performances.

CCS CONCEPTS

• Human-centered computing; • Virtual reality;; • Applied computing; • Performing arts;;

** Place the footnote text for the author (if applicable) here.

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KEYWORDS

Intangible cultural heritage, VR performance, storytelling, interaction design

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1 INTRODUCTION

“Hua’er” is deeply rooted in everyday life of the people in Ningxia. Due to the influence of local dialect, most of the audience cannot directly understand the meaning of the lyrics, and it is difficult to appreciate the emotional and cultural content contained in “Hua’er”. Based on the current trends in VR and gesture recognition technology [Selmanović et al., 2020; Han et al., 2019; Georgiadis and Yousefi, 2017], we recreated the development of “Hua’er”’s heritage and basic repertoire in a virtual environment, allowing the audience to learn about the history, the singing form and the folklife from the perspective of the inheritor of “Hua’er”. We focus on three approaches: interactive performance narrative, metaphorical elements, and embodied cognition. The VR experience, “Flower and the Youth”, presents and interprets the lyrical content and emotional content of “Hua’er” performance and demonstrates the region’s cultural landscape where “Hua’er” is located.

2 INTERACTIVE PERFORMANCE NARRATIVE

In general, the VR performance tend to use non-linear narrative methods for storytelling [Christopoulos et al., 2013]. In "Flower and the Youth", we use a virtual scroll as cues for the non-linear narrative, triggering critical events to participate into the plot. We use Oculus quest 2 gesture recognition technology to apply three hand gestures that form a complete interactive narrative experience. In the first scene, we use spotlights as visual guidance, the audience can point their thumb and index fingers to the target location for teleporting by using distance detection. In the second scene, the virtual avatars surround the campfire hand the audience some flower petals, which the audience can grab directly with their hands and spread their left palms to collect them, completing the narrative process of the flower gathering. In the third scene, the audience pick up and flick through the objects of the working environment (sheet music, letters, old photographs, etc.) to obtain fragments of the "Hua'er" repertoire, helping to inspire the characters and advance the narrative (see Figure 1).

3 METAPHORICAL ELEMENTS

Designing metaphorical performing elements is an important method to create atmosphere, advance the narrative and re-shape the character in intangible cultural heritage performing arts. In this work, we use elements of Ningxia identity such as flowers and Chinese paper-cutting to construct the virtual environment. Throughout the performance, flowers are used as hand-held objects to connect the three scenes. In each scene, the audience will continuously pick up and collect petals, and the virtual flower in their left hand will change from a bud state to a fully bloomed state, suggesting that the audience complete the experience successfully. The flower also represents the resilience and beauty of women. In the end, the petals of the blossoming flower in the audience's left hand re-float into the air, representing the "Hua'er" being passed on by the character and being widely disseminated. Most scenes use the artistic effect of paper-cutting to show the ecological environment of Ningxia (see Figure 2d.).

4 EMBODIED COGNITION

According to the theory of embodied cognition, the audience perception is closely related to their emotional experience. In the creation of the intangible cultural heritage performing arts, the audience participates the narrative from a first-person perspective, advancing their perception of their own identity and enhancing immersion through representational avatars and intuitive behaviours. In this work, we use various reflecting interface (e.g. mirrors in the scene), as well as interaction with other virtual characters and the environment (e.g. a young girl picking up a handbell and swing, a virtual crowd waving and dancing at a bonfire gathering, a music score on the tabletop of the room), so the audience will gradually perceive the identity of the character they are playing as the inheritors of "Hua'er" (see Figure 2e.).



Figure 2: (d) The metaphorical elements: flowers and Chinese paper-cutting. (e) The audience perceives the identity of the virtual character that he/she is playing.

5 SUMMARY

Most of the audience said that the performance is interesting based on ethnically features with realistic and natural gesture interactions. After the experience, they began to develop an interest in "Hua'er" and were able to retell the story according to their own understandings. By integrating gesture recognition technology with interactive narrative, metaphorical elements and embodied cognition, this work has created a VR experience that allows more audience from different cultural backgrounds to experience the charm of the performing arts of intangible cultural heritage.

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